CSCI 131 — Introduction to Programming in Python
Programming Assignment 10
Functions that Return

There are no starter files. Open IDLE, choose **File** → **New File** for each program, and save it under the name given in the problem. Put your name in a comment on the first line of each file.

Problem 1 — First Match

Write a function `find_first(words, letter)` which returns the first string in the list `words` that begins with `letter`. If no string in the list begins with `letter`, the function returns `None`. Save it as `first.py`.

At the bottom of your file, call it like this:

```
print(find_first(['banana', 'apple', 'avocado'], 'a'))
```

Your program should print these lines, one for each call:

Call	Output
`find_first(['banana', 'apple', 'avocado'], 'a')`	`apple`
`find_first(['Apple', 'avocado'], 'a')`	`avocado`
`find_first([], 'a')`	`None`

Problem 2 — Palindromes

A *palindrome* is a word which reads the same backwards as forwards, like `racecar`. Write a function `is_palindrome(word)` which returns `True` if `word` is a palindrome, and `False` if it is not. Save it as `palindrome.py`.

You must check the word by comparing its letters in pairs, with a loop over positions.

At the bottom of your file, call it like this:

```
print(is_palindrome('racecar'))
```

Your program must print these lines, one for each call:

Call	Output
`is_palindrome('racecar')`	`True`
`is_palindrome('a')`	`True`
`is_palindrome('python')`	`False`
`is_palindrome('abca')`	`False`

Return the values `True` and `False`, not the strings `'True'` and `'False'`.

Submission

Submit your source files (`.py`) on Canvas by the end of the period. Put your name in a comment at the top of each file. If a classmate helped you, put their name in the comment as well.

If you are not finished when the period ends, submit what you have — a partial program earns partial credit, an empty one earns none. You may then complete it on your own and resubmit by Sunday, November 8 at 11:59 pm for up to 50% of the remaining credit.